Telemetry Alerting & Dashboard Ruleset: Detecting Firmware-Induced Degradation

Category: Technical | Model: **VFIP-RS-ALERTS-TELEM** | Purpose: Ruleset and implementation notes for telemetry alerting and dashboard thresholds to detect firmware-induced performance degradation (XG-440 firmware incident)

Table of Contents

- 1. Overview & Purpose
- 2. Alert Logic & Thresholds
  - 2.1 Baseline Deviation Logic (Net Generation)
  - 2.2 Protective Trip Count Logic (TRIP-31, TRIP-44)
  - 2.3 Inverter Availability Rules
  - 2.4 Alert Severity Mapping
- 3. Required SCADA Fields & Data Model
- 4. Retention Windows & Data Aggregation
- 5. Sample Alert Expressions (Telemetry System)
- 6. Dashboard Panel Mapping & Highlighting
- 7. Automated Notification Workflows
- 8. Vendor Firmware Release Notes (v3.14.2 → v3.15.0)
- 9. Vendor Change Ticket Summaries (Rollout & Hotfix)
- 10. O&M Incident Report & Postmortem (XG-440)
- 11. Dispatch Runbooks & CMMS Mapping
- 12. Settlement & Imbalance Charge Guidance (ERCOT)
- 13. Troubleshooting Procedures & Flowcharts
- 14. Appendix: Codes, Field Dictionary, Sample Data

1. Overview & Purpose

This document provides a complete ruleset, implementation notes, and operational procedures to detect and respond to firmware-induced performance degradation affecting distributed inverters running XG-440 family firmware. It is designed for integration into telemetry platforms, dashboards, and operational workflows. The content has been synthesized from the vendor release notes, change tickets, O&M incident and postmortem, and settlement guidance tied to the XG-440 incident period (2025-11 to 2025-12).

Scope: - Detect sustained production deviation from baseline - Detect increases in protective trips (notably TRIP-31, TRIP-44) - Monitor inverter availability thresholds - Provide alert severity mapping, retention windows, dashboard mappings, and

notification workflows - Provide vendor-change context, incident timeline, dispatch runbooks, and settlement reconciliation guidance

2. Alert Logic & Thresholds

2.1 Baseline Deviation Logic (Net Generation)

Objective: detect sustained negative deviation in daily net AC generation attributable to firmware behavior.

Rule Summary

- Trigger when a single asset's rolling 3-day average daily net generation is >10% below the asset's 30-day baseline median for the same hour-of-day window and the deviation persists for at least 48 continuous hours.
- Secondary cluster-level rule: trigger when >3 inverters at a plant show the above condition within 72 hours.

Prerequisites

- Baseline calculation window: last 30 calendar days excluding days with manually annotated curtailment or grid outages.
- Required field: `net_ac_energy_daily_kWh` (daily aggregated)
- Required SCADA timestamp resolution: 1 minute or better for intraday validation.

Verification Steps

1. Verify daily aggregation (00:00–23:59) for `net_ac_energy_daily_kWh`.
2. Recompute 30-day baseline median for matching hour-of-day using historical daily data excluding known events.
3. Compute percentage deviation: (baseline - observed) / baseline * 100.
4. Confirm persistence: deviation >10% for minimum 48 hours (two consecutive daily samples if daily resolution; for intraday evaluation use 48 × 1-hour samples).

Prevention Tips

- Annotate curtailments and known grid events in the telemetry system to avoid baseline pollution.
- Run nightly baseline recalculation jobs with logs to track data exclusions.

2.2 Protective Trip Count Logic (TRIP-31, TRIP-44)

Objective: identify firmware changes that increase frequency of protective trips, which may indicate logic regressions.

Rule Summary

Trip Code	Symptom	Trigger Condition	Escalation
TRIP-31	Protective AC disconnect trip (per vendor definition)	Consecutive increase: >= 3 TRIP-31 events on the same inverter within 24 hours, AND 200% increase vs prior 7-day average	Auto-alert to APM; severity = High if plant-wide rate exceeds 10 events/day
TRIP-44	Grid-safety anti-islanding trip	Any occurrence followed by capacity derate >5% for >1 hour	Escalate to vendor immediately; severity = Critical if persistent

Root-Cause Triage Steps

1. Confirm trip timestamps and compare to instantaneous and rolling RMS voltage/Hz signals in SCADA.
2. Check firmware version on inverter via `fw_version` field; if between v3.14.2 and v3.15.0, flag for vendor correlation.
3. Collect inverter logs for the event window (±30 minutes) using remote log pull.
4. If multiple inverters show identical timing pattern, suspect systemic firmware issue; cross-reference change ticket rollouts.

2.3 Inverter Availability Rules

Objective: detect availability drops that could indicate firmware-induced hangs, reboots, or communication failures.

Rule

Trigger when `inverter_availability_pct` per device falls below 96% over a rolling 7-day window, with a minimum continuous downtime segment > 30 minutes.

Details

- Availability is computed as (minutes_reporting / total_minutes) * 100, where reporting means heartbeat plus valid telemetry packets.
- Exclude scheduled maintenance windows by cross-referencing planned_maintenance calendar field.

Verification

1. Confirm heartbeat metric is present: `last_heartbeat_ts` and `heartbeat_interval_sec`.
2. For devices below 96% check for correlated alarm types: communication lost, restart_count >0, fw_update_in_progress flag.

2.4 Alert Severity Mapping

Mapping used to control notification channels and SLAs.

Severity	Criteria	Notification Channel	SLA (Acknowledge / Mitigate)
	Plant-wide production loss >25% across >50% of inverters; TRIP-44 persistent; settlement exposure confirmed	SMS + Email + Incident Bridge; Vendor paged	30 min / 4 hours
	Per-inverter >10% sustained generation deviation; TRIP-31 bursts across multiple inverters; availability <90%	Email + Ticketing (CMMS); Vendor notified top-of-day	1 hour / 8 hours
Medium	Single inverter deviation 5–10%; availability 90–96%	Email to Asset Performance Analyst	4 hours / 48 hours
Low	Telemetry hiccups, transient errors without production impact	Daily digest	By next business day

3. Required SCADA Fields & Data Model

The ruleset relies on the following canonical fields. Field names are normalized for telemetry platform ingestion.

Field Name	Type	Description	Notes
timestamp	ISO8601	UTC timestamp for sample	1-minute resolution preferred
site_id	string	Plant unique identifier	Must match CMMS site key
inverter_serial	string	Inverter serial number	Vendor format: XG-440-XXXXX
fw_version	string	Firmware version running on inverter	Examples: v3.14.2, v3.15.0
net_ac_power_W	number	Instantaneous net AC power (W)	Required for intraday derate detection
net_ac_energy_daily_kWh	number	Daily aggregated net AC energy	UTC day aggregation

trip_code	string	Protective trip code when event occurs	Examples: TRIP-31, TRIP-44
trip_count_24h	integer	Rolling 24h count of trip events	Computed in ingestion
inverter_availability_pct	number	Availability percentage over rolling window	7-day rolling recommended
restart_count	integer	Number of device reboots in window	Used to detect hangs
last_heartbeat_ts	ISO8601	Most recent heartbeat timestamp	Used to compute availability
curtailment_flag	boolean	Operator curtailment annotation	True if manually curtailed
grid_event_flag	boolean	Utility/ISO grid event annotation	Exclude from baseline

Note: Additional vendor log payloads should be stored in compressed JSON blobs keyed by `inverter_serial` and `timestamp` for forensic retrieval.

4. Retention Windows & Data Aggregation

Retention design balances forensic needs with storage costs. Suggested retention windows:

Data Type	Resolution	Retention	Rationale
Raw telemetry (net_ac_power_W, trip_code, heartbeat)	1 min	90 days	For rapid incident triage and baseline recalculation
Daily aggregates (net_ac_energy_daily_kWh)	daily	3 years	Performance trends and settlement reconciliation
Event logs (vendor logs, trip details)	as emitted	2 years	Root-cause analysis and warranty claims
Baselines & models	model artifacts	5 years	Regulatory audits, RAG retrieval

Aggregation guidance:

- Compute and store rolling statistics (7-day, 30-day median/mean, standard deviation) nightly.
- Store baseline flags indicating exclusions with reasons (e.g., curtailment, grid outage, maintenance).

5. Sample Alert Expressions (Telemetry System)

Below are example expressions formatted for a generic telemetry rules engine (pseudocode compatible with common platforms like Prometheus alerting, TimescaleDB continuous queries, or vendor rule DSLs). Adapt field names to your platform.

5.1 Baseline Deviation — Per-Inverter (Daily)

```
# Pseudocode: Trigger if 3-day rolling average is >10% below 30-day baseline
ALERT Firmware_Degradation_NetGen
IF
  rolling_mean(net_ac_energy_daily_kWh, days=3, filter=(curtailment_flag == 
  <
  0.90 * baseline_median(net_ac_energy_daily_kWh, days=30, by=hour_of_day)
FOR 48h
LABELS {severity="high", category="firmware"}
ANNOTATIONS {
  summary="Sustained >10% deviation from 30-day baseline",
  runbook="See VFIP-RS-ALERTS-TELEM section 13: Troubleshooting"
}
```

5.2 TRIP-31 Burst

```
# Trigger if trip_count_24h >=3 and >=200% vs prior 7-day average
ALERT Trip31_Burst
IF
  trip_count_24h >= 3 AND
  trip_count_24h >= 2 * mean(trip_count_24h, days=7)
FOR 1h
LABELS {severity="high", trip="TRIP-31"}
```

5.3 Availability Drop

```
# Trigger when inverter availability falls below 96% over 7 days
ALERT Inverter_Availability_Drop
IF
  inverter_availability_pct < 96 AND
  rolling_window_duration(inverter_availability_pct, days=7)
FOR 24h
LABELS {severity="medium"}
```

Notes: FOR/for durations depend on your rules engine. Use cardinality controls to suppress noisy alerts (e.g., mute if invenoter is under maintenance).

6. Dashboard Panel Mapping & Highlighting

When an alert fires, dashboards should highlight relevant signals and provide

context for rapid triage. Suggested panel mappings:

Alert	Primary Panels	Signals to Highlight	Context Widgets
Firmware_Degradation_NetGen	Plant Production Trend, Inverter Comparison	`net_ac_energy_daily_kWh`, `net_ac_power_W`, baseline overlay	Baseline heatmap (30-day), Annotations timeline (curtailments)
Trip31_Burst	Alarm Timeline, Trip Event Log	`trip_code`, `trip_count_24h`, instant volt/freq profiles	Vendor firmware map (which inverters on v3.14.2–v3.15.0)
Inverter_Availability_Drop	Availability Heatmap, Device Status Grid	`inverter_availability_pct`, `last_heartbeat_ts`, restart_count	Contact list, Last maintenance record

Dashboard Visual Cues

- Use red outlines for Critical panels, orange for High, yellow for Medium.
- Display firmware version column; allow filtering by version range (e.g., v3.14.2→v3.15.0).
- Include quick action buttons: "Open Ticket", "Request Logs", "Run Remote Diagnostics".

7. Automated Notification Workflows

Automated workflows reduce time-to-acknowledge. The following workflows cover notification steps for each severity and include vendor contact integration.

7.1 Critical Workflow

1. Alert triggers → System creates incident in Incident Management (IM) with priority P1.
2. Automated SMS and Email sent to Asset Performance Analyst (APA) on-call + Plant O&M Lead + Vendor Tier-1 contact.
3. System opens vendor portal ticket (if not already) using API: include device list, fw_version, raw trip logs (zipped), and baseline deviation plots.
4. Auto-create Incident Bridge and invite vendor, APA, Dispatch, and Regulatory contact if settlement exposure is detected.

5. Ticket contains required attachments: 7-day rolling telemetry, sample raw logs ±1 hour, plant-level generation summary.

7.2 High/Medium Workflow

1. Create CMMS corrective work order when physical site visit is required. Template populated using dispatch runbook (see section 11).
2. Email to APA and vendor support with SLA expectations (see severity mapping).
3. Automated daily digest summarizing alert status until resolved.

7.3 Vendor Contact & Escalation Matrix

Role	Name (Example)	Contact Method	Escalation Time
Vendor Tier-1	VendorSupport@vendor.com	Email	Immediate
Vendor On-Call Engineer	+1-555-0101	Phone/SMS	30 min for Critical
Vendor Engineering Lead	eng-lead@vendor.com	Email	2 hours

All outgoing messages should include correlation IDs, list of impacted inverter_serials, firmware versions, and ticket ID for traceability.

8. Vendor Firmware Release Notes (v3.14.2 → v3.15.0)

These vendor-provided notes have been synthesized to capture parameter changes, protective trip behaviors, and rollback conditions relevant to the XG-440 incident.

v3.14.2 (Released 2025-11-15)

- Change: Modified anti-islanding timeout to improve fault response. Parameter: `AI_timeout_ms` default changed from 150 to 120.
- Change: Added adaptive derate logic under high reactive demand. Parameter: `adaptive_derate_enable` = true.
- Notes: TRIP-31 behavior unchanged formally, but event logging format updated (`trip_detail_v2`).
- Rollback conditions recommended: if TRIP-31 occurrences increased by >100% in 24h post-rollout.

v3.14.8 (Intermediate patch - Released 2025-11-20)

- Change: Adjusted reactive current threshold for LVRT module. Parameter:

`LVRT_reactive_threshold_pct` set to 18% (from 22%).
- Reported issue: under certain grid profiles this patch increased TRIP-31 occurrences due to mis-evaluated voltage sag window.
- Rollback recommendation: revert to v3.14.2 if TRIP-31 occurrences exceed 3 per inverter per day.

v3.15.0 (Released 2025-12-08 — Hotfix)

- Change: Fix for voltage-sag detection window with corrected hysteresis. Parameter: `voltage_sag_hysteresis_ms` increased from 20 to 40ms.
- Fix: Resolved memory leak in trip logging module; reduced log emission frequency for non-fatal events.
- New telemetry: `trip_log_hash` added to correlate events across fleet.
- Rollback guidance: If inverter availability drops >5% after upgrade or if TRIP-44 occurs after nightly grid events, immediately initiate rollback per vendor instructions.

Important: The vendor notes explicitly call out TRIP-31 and TRIP-44 as the primary protective trips modified across these releases. Use firmware mapping to quickly identify at-risk devices.

9. Vendor Change Ticket Summaries (Rollout & Hotfix)

Summaries of change tickets logged in vendor portal during the rollout window.

Rollout 2025-11-20 → 2025-11-21 (Ticket: VEND-RT-20251120)

Scope:

- Planned rollout of v3.14.8 to production fleet subset: inverter_serial ranges XG-440-10000 → XG-440-19999 and XG-440-30000 → XG-440-34999.
- Plants impacted: SiteA (100 inverters), SiteB (75 inverters), SiteC (40 inverters).

Notes:

- Change window: 2025-11-20 01:00–05:00 UTC.
- Rollback condition: automatic rollback flag set if TRIP-31 count per device >=3 within 24h post-update.
- Change ticket attachments: update manifest, pre/post-update checksum, expected telemetry delta report.

Hotfix 2025-12-08 (Ticket: VEND-HF-20251208)

Scope:

- Emergency hotfix to v3.15.0 deployed to all devices that had v3.14.8 applied between 2025-11-20 and 2025-11-21.

- Target serial ranges same as rollout; additional high-risk devices: XG-440-20001→XG-440-20500.

Notes:

- Change window: 2025-12-08 00:30–03:30 UTC.
- Vendor recommended immediate log collection pre/post hotfix; provided script `collect_fw_logs.sh` executed via vendor remote agent.

10. O&M Incident Report & Postmortem (XG-440)

This section summarizes the on-site and remote operations timelines, curtailment windows, and validation steps performed by O&M during the incident.

Timeline (Day-by-Day)

Date (UTC)	Summary	Key Actions
2025-11-20	Planned rollout of v3.14.8 initiated. Initial telemetry showed slight uptick in trip logs at SiteA during morning ramp.	Vendor and APA monitored. Collected baseline snapshots pre-update.
2025-11-21	Multiple TRIP-31 events observed across SiteA and SiteB. Net generation deviation on several inverters >10% vs 30-day baseline.	Opened internal incident (INC-20251121-XG). Initiated log retrieval and placed tentative hold on further rollouts.
2025-11-22	Understood correlation with v3.14.8. Vendor recommended targeted rollback for affected serial ranges.	Executed rollback for top 10 impacted inverters at SiteA. Observed production recovery in 2–4 hours for rolled-back units.
2025-11-23 → 2025-12-07	Ongoing monitoring; staged vetting of vendor hotfix.	Collected additional telemetry and compared to control group inverters on v3.14.2.
2025-12-08	Vendor released hotfix v3.15.0 and issued immediate deployment to impacted range.	Executed hotfix. Close monitoring resumed. TRIP rates reduced but availability issues emerged for subset.
2025-12-09 → 2025-12-15	Post-hotfix monitoring and final remediation of firmware regressions.	Final rollback performed for devices with persistent availability drop; coordination with vendor for patch updates.

Curtailment Window Descriptions

During incident operations, several manual curtailment windows were executed to

prevent nuisance tripping and minimize settlement exposure. Curtailment windows are annotated in telemetry with `curtailment_flag` and a `curtailment_reason` field.

- 2025-11-21 14:00–18:00 UTC — Manual derate at SiteA (-20%) to reduce trip incidence. Reason: reduce AC reactive load on problematic feeders.
- 2025-11-22 09:00–11:30 UTC — Short isolations on specific inverter blocks for rollback.
- 2025-12-08 01:00–03:00 UTC — Temporary curtailment at SiteB during hotfix deployment.

Validation Steps Performed

1. Comparison of per-inverter 30-day baseline to post-update 7-day median.
2. Trip log correlation across devices using `trip_log_hash` introduced in v3.15.0.
3. Sample log extraction and vendor-assisted debugging to isolate voltage-sag detection misclassification.
4. Controlled rollback and observation of recovery trajectories (time-to-recover metrics recorded).

11. Dispatch Runbooks & CMMS Mapping

Runbooks map SCADA alarms to CMMS corrective work orders and provide step-by-step technician instructions.

Runbook Example: TRIP-31 Burst at Single Inverter

1. Triage: Confirm trip events via telemetry and collect `trip_detail_v2` payloads.
2. Check firmware: if `fw_version` in [v3.14.2, v3.14.8] mark as candidate for firmware regression.
3. Create CMMS Work Order Template: WO-TYPE = "Electrical - Inverter Trip", Priority = High, Tasks:

    1. Remote log pull: `collect_fw_logs.sh --serial {inverter_serial} --window {t0-30m}:{t0+30m}`
    2. Perform remote health check (heartbeat, restart_count)
    3. If reboot_count > 2, schedule site visit
    4. If firmware version is vulnerable and rollback authorized, coordinate rollback with vendor

4. Attach telemetry snapshots and vendor ticket number to work order.

CMMS Work Order Template Fields

Field	Value / Example
WO_ID	Auto-generated (e.g., WO-20251121-0001)
Site	SiteA
Asset	Inverter XG-440-10123
Priority	High
Tasks	Remote log, reboot, visual inspection, firmware rollback
Attachments	Telemetry snapshots, vendor ticket link
Resolution Codes	FW_ROLLBACK, REMOTE_REBOOT, SITE_VISIT, NO_FAULT_FOUND

12. Settlement & Imbalance Charge Guidance (ERCOT)

Firmware-induced derates can create imbalance exposures. Guidance below explains how penalties are calculated and what fields to reconcile when investigating settlement impacts.

ERCOT Settlement Overview

Imbalance charges are assessed based on the difference between scheduled or expected generation and actual delivered generation during market settlement intervals (typically 5-minute intervals for ERCOT). Key steps for reconciliation:

1. Collect interval-level actual generation: `net_ac_power_W` sampled at 5-minute intervals or aggregated to MWh per settlement interval.
2. Determine expected/scheduled generation used for settlement (from SCADA/Market data): typically plant-level schedule.
3. Calculate deviation per interval: schedule - actual.
4. Multiply deviation by settlement price for interval to compute imbalance charge.

Fields to Reconcile

Field	Description	Why it's important
interval_ts	Settlement interval timestamp	Aligns measurements to market windows
scheduled_mw	Market schedule (MW)	Baseline for imbalance calculation
actual_mw	Measured delivery (MW)	Used to compute deviation
curtailment_flag	Manual curtailment annotation	May be considered in disputes

event_annotation	Incident ID or vendor ticket	Supports dispute evidence

Settlement Dispute Steps

1. Collect interval-level telemetry and tag intervals with incident metadata (INC ID, firmware version, curtailment).
2. Compute incremental exposure: sum((scheduled_mw - actual_mw) × price) across impacted intervals.
3. Prepare dispute package: include raw SCADA intervals, baseline calculations, vendor log exports, and correlating alarms.
4. Submit to ERCOT per standard dispute procedures within the allowable window, attaching the incident postmortem as supplementary evidence.

Tip: Maintain a "settlement-ready" data export job that can assemble required fields into a single CSV for rapid disputes.

13. Troubleshooting Procedures & Flowcharts

Step-by-step procedures to diagnose and remediate common symptoms associated with firmware-induced degradation.

Procedure: Investigate Sustained Net Generation Deviation

1. Confirm alert: collect alert context including site_id, inverter_serials, timestamps, and severity.
2. Validate telemetry: ensure no gaps in `net_ac_power_W` and `net_ac_energy_daily_kWh` for the window.
3. Check external factors: weather (irradiance), curtailment_flag, grid_event_flag. Exclude natural causes.
4. Check firmware versions: filter inverters to see if those on v3.14.2→v3.15.0 are overrepresented.
5. Correlate trips and restarts: inspect `trip_code` and `restart_count`.
6. If pattern matches firmware regression, open vendor ticket with collected logs and request rollback or hotfix.

Flowchart: High-Level Triage

1) Alert received → 2) Is it curtailment or grid event? (Yes: annotate and close) (No: continue) → 3) Are affected devices on suspect fw versions? (Yes: collect logs, open vendor ticket) → 4) Is there settlement exposure? (Yes: prepare dispute package) → 5) Implement remediation (rollback, hotfix, configuration change) → 6) Monitor recovery and close incident.

Error Codes & Resolution Table

Error Code	Symptom	Root Causes	Resolution

ERR-TRIP31-001	Frequent TRIP-31 events	Firmware LVRT threshold misconfiguration; noisy voltage sags	Collect logs, request vendor patch or rollback to v3.14.2; if immediate, isolate by disabling adaptive derate parameter remotely if safe
ERR-AVAIL-LOW	Availability < 96%	Firmware hang/reboot loop; comms failure	Pull logs, check restart_count; if restart_count high, schedule site visit and rollback candidate
ERR-LOGSYNC-001	Missing vendor logs	Agent not running post-update	Trigger remote agent restart; if not recovered, escalate to vendor and raise CMMS work order

14. Appendix: Codes, Field Dictionary, Sample Data

14.1 Protective Trip Codes

Code	Name	Description
TRIP-31	AC Disconnect Trip	Tripping due to anti-islanding/AC fault conditions
TRIP-44	Grid Safety Anti-Islanding	Protective islanding detection trip

14.2 Field Dictionary (Selected)

Field	Type	Format / Units
net_ac_power_W	number	Watts (instantaneous)
net_ac_energy_daily_kWh	number	kWh (daily aggregated)
trip_code	string	e.g., TRIP-31
fw_version	string	e.g., v3.14.8
inverter_availability_pct	number	percent over rolling window
trip_log_hash	string	Hash correlator introduced in v3.15.0

14.3 Sample CSV Extract (Interval Data)

```
interval_ts,site_id,inverter_serial,fw_version,scheduled_mw,actual_mw,trip_co
2025-11-21T14:05:00Z,SiteA,XG-440-10123,v3.14.8,0.05,0.03,TRIP-31,false
2025-11-21T14:10:00Z,SiteA,XG-440-10123,v3.14.8,0.05,0.02,TRIP-31,false
2025-11-21T14:15:00Z,SiteA,XG-440-10123,v3.14.8,0.05,0.02,TRIP-31,false
```

14.4 Example Vendor Log Retrieval Command

```
# Example script invocation (from vendor portal docs)
collect_fw_logs.sh --serial XG-440-10123 --from "2025-11-21T13:30:00Z" --to "
```

14.5 Quick Reference: Actions By Alert

Alert	Immediate Action	Owner
Firmware_Degradation_NetGen	Collect baseline diffs, open INC, notify vendor	Asset Performance Analyst
Trip31_Burst	Pull trip logs, compare to firmware map, escalate to vendor	Plant O&M Lead
Inverter_Availability_Drop	Check heartbeat, schedule site visit if restart_count high	Field Technician